SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Formal Scientific/Engineering Contract

Scientific contract version: **v0.1**
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Scientific owner: Grant Wilson
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Authority boundary. This is the normative formal view of the implementation-independent scientific contract. It preserves the canonical equations, 52 requirements, 25 predictions, exact state and support semantics, provenance, and conformance routing. The separate science-team rationale r0.7.1 explains the same science without replacing this authority. This document does not claim implementation conformity, representation fidelity, scientific validation, observational performance, or production readiness.

Physical model and contract at a glance

MAP consumes the realized MAP-facing output of PTC; it never bypasses PTC to consume CAL output directly. ALIGN/AST supplies the coordinate facts and MAP supplies the target grid and declared projection. SCI-MAP applies one fixed, positive-coefficient linear averaging operator to the signal and to every declared linear companion. A coadd then applies a second positive-coefficient operator to atomically admitted, compatible observation bundles on the same grid. All scientific statements are conditional on the complete admitted calibration, alignment, eligibility, coefficient, response, geometry, and covariance state.

Facet	Compact authority
Input	Realized SCI-PTC v0.1/r0.5 calibrated-x-derived nonpolarimetric total-intensity-equivalent product in `mJy/beam`, with exact originating nominal-beam identity, occurrence/generation, retention, causes, and response state; exact AST RTC-grid coordinates for the same stable sample and parent; and a future owner-selected PTC MAP coefficient.
Output	A complete immutable observation-map bundle and, when requested and admissible, a complete centered-integer common-grid observation coadd bundle.
Equation	On admitted membership, $N_p = \sum_i a_{pi} z_i$ and $Q_p = \sum_i a_{pi}$. The normalized scientific vector contains exactly the rows authorized by the effective support policy, with $\widehat{m}_p = N_p/Q_p$ there; response and covariance use the same explicit row selection. Coadd repeats this restricted structure over complete observations.
Source	Frozen SCI-PTC v0.1/r0.5 digest `8357961a49272adc40e27a8aa9e760e0 d01ff2419ae2c88a62c0f93c9f959e66`; SCI-CAL v0.1 science r0.5/engineering r0.4 digest `413426f49edf1249f751a05bb8c6e9fd907b11e8da0530fe2da39814885efb22`; SCI-AST v0.1/r0.3 manifest digest `b54b6013750540f28aad02339a60bf36078980dc53b132be ab73069d66ef3601`; SCI-VAL Core v0.1/r0.3 digest `2fc3b3ad329fe303 5d442b43d1e564a74fc86ab49f85f56e87322d8553fad9a6`; `SCI-PTC_TO_SCI-MAP v0.1/r0.1`; and the exact r0.7.1 MAP profile, boundary, owner-decision, and source-binding records.
Status	Contract author draft. Algebraic predictions are preregistered below; none has been executed here. All stronger claim layers remain unassessed.

Contents

1 Scientific boundary and estimand

The ordinary observation map estimates the coefficient-weighted average of the admitted realized PTC sample quantity in each declared map pixel. Its meaning is inherited through the required CAL-to-PTC route, from ALIGN/AST's coordinate relation, and from PTC's transformed signal, product identity, availability, producer-local validity, and MAP-facing coefficient. SCI-MAP does not repair or reinterpret those facts. VAL may register and evaluate a named rule, but it is not the author of producer or MAP policy.

Let i bind observation, detector occurrence and UID, stable RTC output sample n, exact PTC product/application generation, segment, and array/network/group; time is an attribute and storage positions are locators. Here $z_i = Z_i^{\mathrm{PTC}}$ is the realized PTC transformed signal and γ_i is its exact PTC-owner-selected MAP-facing coefficient. The paired x/r symbols remain KID readout coordinates. The effective coefficient is $a_{pi} = G_{pi}\gamma_i$. No MAP-facing coefficient family is selected by frozen SCI-PTC v0.1/r0.5 or bound in the SCI-MAP r0.7.1 source packet. MAP does not select this PTC-owned family. The ordinary numerical route is therefore unavailable rather than defaulting to unity, loading, sensitivity, scatter, or precision.

The fixed contribution set $\mathcal{C}_p$ factors transformed-signal availability, PTC retention, coefficient availability, coefficient/QC, `SCI-MAP:map_upstream_admission@2`, AST validity, finite signal, finite positive coefficient, admitted projection/boundary, and MAP-local companion gates. Each retained typed state σ_g is projected to a Boolean membership predicate b_g; T/F/U/C and VAL's four axes are never conjoined directly. Evaluation follows the structural, safe numerical-classification, coordinate/product, membership, and accumulation stages. Only structurally admitted payloads are read for MAP classification, and all required gates pass before payload arithmetic. Every source state, immutable reason, cause, provenance, and failure scope is retained. Let Π_{eff} be the recorded effective product support policy and $\mathcal{S}_{\mathrm{out}}(\Pi_{\mathrm{eff}})$ the exact set of map rows for which every support predicate that policy designates applicable passes. This is a row domain, not a numerical mask. The frozen ordinary result is then reused directly:

$$a_{pi} = G_{pi}\,\gamma_i, \tag{1}$$

$$\mathcal{G} = \{Z_{\mathrm{avail}}, \mathrm{PTC}_{\mathrm{ret}}, \gamma_{\mathrm{avail}}, \gamma_{\mathrm{QC}}, \mathrm{MAP}_{\mathrm{adm}}, \mathrm{AST}_{\mathrm{coord}}, Z_{\mathrm{finite}}, \gamma_{>0}, G_{\mathrm{place}}, \mathrm{MAP}_{\mathrm{local}}\}, \tag{2}$$

$$b_g(i,p) = \mathrm{pass}_g(\sigma_g(i,p)) \in \{0,1\}, \qquad g \in \mathcal{G}, \tag{3}$$

$$\mathrm{pass}_{\gamma,\mathrm{QC}}(i) = \mathbf{1}\begin{bmatrix} \mathrm{request}_{\gamma,\mathrm{QC}}(i) = \texttt{requested}, \\ \mathrm{applicability}_{\gamma,\mathrm{QC}}(i) = \texttt{applicable}, \\ \mathrm{eligibility}_{\gamma,\mathrm{QC}}(i) = \texttt{eligible}, \\ \mathrm{realization}_{\gamma,\mathrm{QC}}(i) = \texttt{realized} \end{bmatrix}, \tag{4}$$

$$\mathcal{C}_p = \left\{ i : \bigwedge_{g\in\mathcal{G}} b_g(i,p) = 1 \right\}, \tag{5}$$

$$\mathcal{S}_{\mathrm{out}}(\Pi_{\mathrm{eff}}) = \{p : \mathrm{authorize}_{\Pi_{\mathrm{eff}}}(p) = 1\}, \tag{6}$$

$$N_p = \sum_{i\in\mathcal{C}_p} a_{pi} z_i, \qquad Q_p = \sum_{i\in\mathcal{C}_p} a_{pi}, \tag{7}$$

$$K_p = \sum_{i\in\mathcal{C}_p} a_{pi} k_i, \tag{8}$$

$$\widehat{m}_p = N_p/Q_p, \qquad \widehat{k}_p = K_p/Q_p, \qquad p \in \mathcal{S}_{\mathrm{out}}, \tag{9}$$

The ten projections in Equation (3) are exact:

Pass predicate	Owner and object	Retained source codomain	Sole passing state	Nonpass and scope
$b_{Z_{\mathrm{avail}}}(i)$	PTC; transformed occurrence	signal availability with cause	exact `available` Z_i^{PTC}	unavailable, invalid, not requested, or conflict; occurrence/route exclusion

$b_{\mathrm{PTC}_{\mathrm{ret}}}(i)$	PTC policy and VAL; occurrence	named request, applicability, eligibility, and realization fields	requested, applicable, `eligible`, realized	every other field combination; occurrence/route scope
$b_{\gamma_{\mathrm{avail}}}(i)$	PTC; coefficient slot/generation	family/generation, index/broadcast, product compatibility, readable payload, typed state/cause	exact compatible typed payload available; no finiteness claim	missing/unreadable, unavailable, conflict, or generation mismatch; route exclusion
$\mathrm{pass}_{\gamma\mathrm{QC}}(i)$	PTC policy owner; coefficient occurrence	named request, applicability, eligibility, and realization fields plus causes	request=`requested`; applicability=`applicable`; eligibility=`eligible`; realization=`realized` under exact PTC profile	any other named-field combination is nonpassing; structural source-binding failure only when separately present; never MAP admission
$b_{\mathrm{MAP}_{\mathrm{adm}}}(i)$	MAP policy and VAL; occurrence	named request, applicability, eligibility, and realization fields	requested, applicable, `eligible`, realized	every other field combination; occurrence exclusion or source-binding block
$b_{\mathrm{AST}_{\mathrm{coord}}}(i)$	AST; same-n coordinate parent	availability, validity, frame, parent, cause	exact coordinate available and valid on matching parent	invalid, unavailable, conflict, or parent mismatch; occurrence exclusion
$b_{Z_{\mathrm{finite}}}(i)$	MAP numerical gate; z_i	finite classification	finite	NaN, Inf, or unrepresentable; occurrence/product scope
$b_{\gamma>0}(i)$	MAP numerical gate; γ_i	positive; zero; negative; non-finite; unrepresentable class	finite and strictly positive	zero noncontributing; negative, non-finite, or unrepresentable invalid; occurrence scope
$b_{G_{\mathrm{place}}}(i,p)$	MAP; occurrence–pixel relation	one-hot placement/boundary state	unique in-grid half-open containing pixel	invalid coordinate, outer-upper boundary, or out of grid; pixel nonmembership
$b_{\mathrm{MAP}_{\mathrm{local}}}(i,p)$	MAP; product row and role	local and companion decisions	every applicable local predicate passes	false, unknown, conflict, unavailable, or missing companion; declared scope

Boolean zero here means estimator nonmembership only. It is never multiplied into z_i, and it never erases the retained typed record or cause.

The supported division is not a recipe for filling unsupported pixels. Pre-normalized accumulators and representation-specific storage slots may exist outside $\mathcal{S}_{\mathrm{out}}$, but the normalized scientific vector has no row there. A dense zero, sentinel, or coincidentally finite payload is not a semantic substitute and is not measured zero sky. Likewise, Q_p is a normalization fact. Its dimensions or historical name do not make it precision.

The active role is a calibrated-x-derived nonpolarimetric total-intensity-equivalent quantity in `mJy/beam`. The exact originating fixed-nominal-beam/template and CAL/PTC lineage are part of the quantity identity. Neither the unit nor a STOKES token promotes it to Stokes I. V0.1 does not authorize polarimetry, measured-R mapmaking, JINC, maximum-likelihood mapmaking, mosaicking, or reprojection.

2 Linear operator, response, and absolute mode

For the selected frozen plan Π, transformed-sample input domain $\mathcal{I}_\Pi$, projection matrix G, diagonal coefficient matrix Ω, neutral full-grid pixel vector $\boldsymbol{v}$, exact fixed-state upstream-to-PTC response $H_{\text{fixed},\Theta}$, and output-row selector J_{out}, the map operator, fixed-state response, and conditional covariance are:

$$D_Q = \operatorname{diag}(G\Omega\mathbf{1}), \qquad J_{\text{out}}\boldsymbol{v} = (v_p)_{p\in\mathcal{S}_{\text{out}}}, \qquad D_{Q,\text{out}} = J_{\text{out}} D_Q J_{\text{out}}^{\mathsf{T}}, \tag{10}$$

$$A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1} J_{\text{out}} G\Omega, \qquad \widehat{\boldsymbol{m}}_{\text{out}} = A_{\text{MAP},\Pi}\boldsymbol{z}, \tag{11}$$

$$\delta\boldsymbol{m}_{\text{out}} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta}\,\delta\boldsymbol{s}, \qquad R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta}, \qquad C_{m,\text{out}} = A_{\text{MAP},\Pi} N_{\text{PTC}} A_{\text{MAP},\Pi}^{\mathsf{T}}. \tag{12}$$

The identity $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1} J_{\text{out}} G\Omega$ binds $\mathcal{I}_\Pi$, output-row domain $\mathcal{S}_{\text{out}}$, exact coefficient family/generation, `SCI-MAP:one_hot_containing_pixel@1`, support policy, target WCS/grid, immutable parents, and lifecycle generation. The diagonal $D_{Q,\text{out}}$ contains only finite positive normalizations on $\mathcal{S}_{\text{out}}$, so its ordinary inverse is defined. The matrix has no rows outside that set. Extending it to the full storage grid by writing zeros would be a representation convenience, not this scientific operator, response, or covariance.

Here N_{PTC} is a declared covariance of PTC-domain processed-sample error when such information is available, not the ensemble covariance of sky plus error. The state Θ contains all selected or estimated calibration, alignment, membership, coefficients, response, geometry, identity, and effective-support facts. Equation (12) therefore makes a conditional perturbation statement on exactly the authorized rows; it makes no unnamed μ_0 or residual-bias claim. If any element of Θ is estimated from the same data, an unconditional claim requires its joint uncertainty and cross-covariances.

The linear object $H_{\text{fixed},\Theta}$ has an exact source domain, PTC-sample codomain, unit transformation, parent, membership, support, coefficient state, and response basis. It is not a name for a refitted procedure. When available, $R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta}$ describes the response of that exact frozen state on the authorized output domain. A stored kernel image is the realized response to a named unit-bearing template $\boldsymbol{\tau}$ only if its sample-domain parent is $\boldsymbol{k} = H_{\text{fixed},\Theta}\boldsymbol{\tau}$ and the stored map is $A_{\text{MAP},\Pi}\boldsymbol{k} = R_{\text{fixed},\Theta}\boldsymbol{\tau}$ with the same exact row identity as signal. Without that identity it is a tracer, not the measured response. If any signal member lacks the required exact parent, response is limited or unavailable on the unchanged signal membership; no hidden subset, zero, or identity array can stand in for it. A realized PTC-grid companion begins at MAP and receives $A_{\text{MAP},\Pi}$ exactly once; it is never passed through upstream response again.

PTC full-procedure and PTC+MAP re-resolved responses are bounded finite-difference families, not one H matrix:

$$\Delta\boldsymbol{z}_{\text{PTC-FP}}(\delta\boldsymbol{s};\alpha) = \mathcal{P}_{\text{PTC}}(\boldsymbol{s}+\delta\boldsymbol{s};\alpha) - \mathcal{P}_{\text{PTC}}(\boldsymbol{s};\alpha), \tag{13}$$

$$\Delta\mathcal{S}_{\text{PTC-FP}} = \text{state}_{\text{PTC}}(\boldsymbol{s}+\delta\boldsymbol{s};\alpha) \ominus \text{state}_{\text{PTC}}(\boldsymbol{s};\alpha), \tag{14}$$

$$\Delta\boldsymbol{m}_{\text{PTC-FP}} = A_{\text{MAP},\Pi}\,\Delta\boldsymbol{z}_{\text{PTC-FP}}, \qquad \text{only when MAP state } \Pi \text{ is held fixed}, \tag{15}$$

$$\Delta\boldsymbol{m}_{\text{PTC+MAP-RR}} = \mathcal{M}(\mathcal{P}_{\text{PTC}}(\boldsymbol{s}+\delta\boldsymbol{s};\alpha);\Pi_+) - \mathcal{M}(\mathcal{P}_{\text{PTC}}(\boldsymbol{s};\alpha);\Pi_0)\,, \tag{16}$$

Here α binds amplitude, perturbation side, source state, location, domain, and every other family parameter. $\Delta\mathcal{S}_{\text{PTC-FP}}$ separately records changed masks, groups, rank, loading subspace, classifications, coefficient state, stop cause, support, and other selected state. Equation (15) applies only when MAP membership, coefficient, projection, and support remain fixed. Equation (16) instead uses separately resolved Π_0 and Π_+; it is the PTC+MAP re-resolved procedure response, not the fixed MAP operator applied to one matrix. The reserved whole-chain RTC-to-CAL-to-PTC-to-MAP response would rerun RTC, CAL, AST-dependent selection, PTC, MAP admission, coefficient/support resolution, and MAP from exact immutable parents under separate authority; it is unavailable and not claimed. An unavailable response state does not invalidate the numerical map or prohibit later analysis. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity; it does not rewrite the original product's claim.

V0.1 fixes the signal-centering operator to $L = I$. SCI-MAP therefore does not remove a mean or a constant mode during coaddition. It preserves a constant admitted sample vector on the exact support-authorized output rows because numerator and denominator use identical coefficients. A stronger constant-sky statement additionally requires the forward operator to map that sky mode to the admitted sample constant. Neither result alone establishes point-source, integrated-flux, extended-source, edge, or absolute astrometric fidelity. The PTC nonrestored additive reference λ, fitted correlated removed component $\widehat{U}_{\mathrm{corr}}$, total removed component $U_{\mathrm{total},\Theta}$, removed-subspace identity, fixed-state null space, full-procedure invariant/unidentifiable modes, and any separately authorized residual conditional bias are distinct facts. Any absolute mode already removed upstream remains absent and must be carried in response and null-mode provenance.

3 Input classes and falsifiable response behavior

The same response equation yields distinct, testable input classes without granting them equal scientific meaning:

Constant.
Fixed positive coefficients preserve a constant admitted sample vector even when coverage and variance vary.

Delta or basis pixel.
The output is a support-row-restricted column of $R_{\mathrm{fixed},\Theta}$, including beam, projection, preprocessing, and boundary effects.

Point-source template.
The output is $R_{\mathrm{fixed},\Theta}\boldsymbol{\tau}_b$ on exactly the authorized rows. Peak, integrated, and fit amplitudes are different functionals.

Resolved, gradient, and Fourier structure.
The output is $R_{\mathrm{fixed},\Theta}\boldsymbol{\tau}$; constant preservation says nothing about these modes.

Variable coverage.
Fixed-coefficient normalization may preserve a constant while covariance and nonconstant response vary spatially.

Finite boundary.
Truncation can preserve a local constant yet change a compact-source peak or integral. No wrap or clamp is authorized.

The authorized base family is `SCI-MAP:one_hot_containing_pixel@1`. It uses half-open pixel cells, assigns unity to the unique finite in-grid containing pixel, excludes the outer upper boundary, never wraps or clamps, and has $\sum_p G_{pi} = 1$ only on that in-grid domain. Fractional projection remains a future role; no kernel, conservation, boundary-loss, response, or covariance semantics are inferred for it.

Response fidelity is therefore a validation layer, not a consequence of code executing the algebra. Delta, point-template, extended, Fourier, variable- coverage, and edge injections are preregistered in Appendix D; they are not run or claimed here.

4 Conditional uncertainty and statistical labels

For independent selected sample errors with a future exact coefficient family proven to be true inverse variance, the general formal relation is:

$$\mathrm{Var}(\widehat{m}_p \mid \Theta) = \frac{\sum_i G_{pi}^2 \gamma_i}{(\sum_i G_{pi}\gamma_i)^2}, \qquad \mathrm{Cov}(\widehat{m}_p, \widehat{m}_r \mid \Theta) = \frac{\sum_i G_{pi} G_{ri} \gamma_i}{Q_p Q_r}, \qquad p, r \in \mathcal{S}_{\mathrm{out}}, \tag{17}$$

$$\phi_p = \frac{Q_p^2}{\sum_i G_{pi}^2 \gamma_i}, \qquad \phi_p = Q_p \Longleftrightarrow \sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0, \qquad p \in \mathcal{S}_{\mathrm{out}}, \tag{18}$$

Equation (18) states why normalization is not precision by default. One-hot geometry satisfies the projection part of the equality, but cannot establish that γ is calibrated inverse variance or that sample errors are independent. Frozen PTC r0.5 has selected no γ family.

Four uncertainty objects must not be conflated:

1. Q, an ordinary normalization with its declared gridding unit;
2. $\phi = 1/[C_{m,\mathrm{out}}]_{pp}$, a conditional marginal inverse variance under its named assumptions;
3. $C_{m,\mathrm{out}}$ or $C_{m,\mathrm{out}}^{+}$, the full conditional covariance or precision on the identifiable stochastic subspace of the exact authorized row domain; and
4. an empirical covariance, weight, or significance calibrated by NOI.

The bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. Missing correlations are named, not set to zero. Neither normalization, hits, exposure, nor diagnostic weight is silently promoted to covariance. Absence of a complete covariance model does not invalidate the map or prohibit later scientific analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity. The minimum persistence form remains an owner decision.

5 Distinct map facts, support, and MAP-local validity

The base/unfiltered MAP bundle separates facts because they answer different questions:

Fact	Meaning
AST-geometric incidence	Which coordinate-valid occurrences touch the unique containing pixel, independent of signal availability.
MAP-route-candidate incidence	Which exact realized PTC occurrences also pass the occurrence-level MAP upstream-admission profile.
Estimator-admitted contributions	How many occurrences passed every ordinary coefficient and contribution predicate.
Admitted observation count	How many complete observation bundles contribute to a coadd pixel; not sample hits or attempted observations.
Upstream-eligible original-footprint exposure	Unique original acquisition footprints ancestral to route candidates, placed by each original's own AST ALIGN-grid coordinate in the target WCS; not complete temporal support.
Retained original-footprint exposure	The same original-footprint estimand restricted to ancestors of estimator-admitted membership; not precision or effective integration time.
Normalization support	The numerical threshold predicate used to authorize ordinary division.
Science-policy support	The separately stricter adopted policy predicate.
MAP-local base-product validity	MAP's conjunction of admitted identity, support, finite signal, and no MAP-product failure; not downstream eligibility.

No one fact is an alias for another. In particular, exposure is not precision, positive normalization is not final validity, and a finite payload is not evidence of scientific availability.

ALIGN owns e_{acq} and valid-original e_{vo}, in seconds on stable original occurrences. RTC, CAL, and PTC preserve those facts, exact lineage, and causes. MAP deduplicates by observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation. Under `SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1`, each positive-e_{vo} original is placed exactly once using its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, with exact geometry/pointing realization, frame, validity, boundary, parent, and version. A descendant RTC-output coordinate is never substituted:

$$\mathcal{U}_p^{\mathrm{up}} = \left\{u : e_u^{\mathrm{vo}} > 0,\ G_p^u = 1,\ \exists i \in \mathcal{I}_{\mathrm{route}} : u \rightsquigarrow i\right\}, \tag{19}$$

$$E_p^{\mathrm{up,footprint}} = \sum_{u \in \mathcal{U}_p^{\mathrm{up}}} e_u^{\mathrm{vo}}, \tag{20}$$

$$\mathcal{U}_p^{\mathrm{ret}} = \left\{u : e_u^{\mathrm{vo}} > 0,\ G_p^u = 1,\ \exists i \in \mathcal{C} : u \rightsquigarrow i\right\}, \qquad E_p^{\mathrm{ret,footprint}} = \sum_{u \in \mathcal{U}_p^{\mathrm{ret}}} e_u^{\mathrm{vo}}, \tag{21}$$

$$\mathcal{U}_p^{c} = \bigcup_{o \in \mathcal{O}_p} \{(o,u) : u \in \mathcal{U}_{op}^{\mathrm{ret}}\}, \qquad E_p^{c} = \sum_{(o,u) \in \mathcal{U}_p^{c}} e_{ou}^{\mathrm{vo}}. \tag{22}$$

The relation $u \rightsquigarrow i$ is causal influence only. One original may influence several RTC/PTC descendants at different MAP pixels, but those edges do not relocate or duplicate its physical detector-seconds. Under the base one-hot family an original contributes exposure to at most one pixel, or none after boundary loss. Upstream-eligible exposure uses originals ancestral to the exact route-candidate population; retained exposure restricts that population to ancestors of estimator-admitted outputs. Coadd identity includes the observation, so union never merges distinct acquisitions. Overlapping filters, decimation, replacement, synthesis, donor reuse, cadence, hits, Q, and γ create no exposure. Missing exact original lineage or own-coordinate authority makes the affected exposure role unavailable rather than reconstructible from duration or count. Causal influence/support is published separately from representative RTC occurrence and physical original-footprint exposure. Neither exposure plane is complete temporal support, effective map integration time, precision, or normalized-map influence.

The adopted v0.1 threshold rule is policy authority, not a derived physical law. For the declared plane and policy population of finite positive Q_p values, sorted in ascending order, the admitted formula names $c = \texttt{coverage\_cut}$:

$$\mathcal{I}_{\Pi_{\mathrm{eff}}} = \text{the declared support-policy population,} \tag{23}$$

$$\mathcal{P} = \{Q_p : \mathrm{finite}(Q_p),\ Q_p > 0,\ p \in \mathcal{I}_{\Pi_{\mathrm{eff}}}\}, \qquad N = |\mathcal{P}|, \tag{24}$$

$$k = \left\lfloor \frac{\lfloor 0.75N \rfloor + N}{2} \right\rfloor, \qquad Q_\star = \begin{cases} \mathcal{P}_{(k)}, & N > 0, \\ 0, & N = 0, \end{cases} \tag{25}$$

$$c = \texttt{coverage\_cut}\ \text{(dimensionless scalar)}, \qquad \mathrm{admit}_{\Pi_{\mathrm{eff}}}(c) = 1, \tag{26}$$

$$T_{\mathrm{norm}} = Q_\star c/10, \qquad T_{\mathrm{sci}} = Q_\star c, \tag{27}$$

$$S_p^{\mathrm{norm}} = \mathbf{1}[\mathrm{finite}(Q_p),\ Q_p > 0,\ Q_p \geq T_{\mathrm{norm}}], \tag{28}$$

$$S_p^{\mathrm{sci}} = \mathbf{1}[\mathrm{finite}(Q_p),\ Q_p > 0,\ Q_p \geq T_{\mathrm{sci}}]. \tag{29}$$

Both predicates require the pixel's own explicit finite positive coefficient; the empty-population threshold of zero does not manufacture support. The operator row set $\mathcal{S}_{\mathrm{out}}$ is the exact intersection of the predicate row sets designated applicable by Π_{eff}; it is not a full-grid mask with zero semantics.

The scientific owner has fixed $c = \texttt{coverage\_cut}$ as a dimensionless support-policy scalar because Q_p, $Q_\star$, and both thresholds are directly compared in the same coefficient unit. `SCI-MAP-CI-001` records the resolved inconsistency and the bounded amendment to the canonical equations, REQ-031/032, and PRED-012. This disposition does not establish a numerical domain. Negative, zero, non-finite, and greater-than-one values, the recommended range, policy authority, and failure behavior remain SCI-MAP-OD-007. Threshold rationale and future change authority remain SCI-MAP-OD-001 and SCI-MAP-OD-002.

6 Observation coaddition on one common grid

Coaddition consumes immutable complete observation bundles, not independent planes. The MAP-owned aggregate profile `SCI-MAP:observation_coadd_admission@1` is registered and evaluated by SCI-VAL. Its four axes independently record request, applicability, eligibility, and realization. It atomically checks quantity and nominal-beam convention, PTC route/product role, response source/class, complete AST frame/WCS/grid, support policy, coefficient family, covariance disclosure, null/additive-reference state, required companions, lifecycle, and parentage before any coadd-owned state changes.

The only v0.1 placement is centered integer embedding on the same grid. Observation and coadd row and column differences are nonnegative and even; their half-differences are the integer offsets, and the shifted reference pixels identify the same world coordinate. The lossless typed WCS is the authority. No fractional shift, interpolation, reprojection, wrapping, or implicit source recentering is allowed.

For the atomically admitted per-pixel set $\mathcal{O}_p$, every contributing observation row is already a member of that observation's support-authorized scientific domain. Let $\mathcal{S}^c_{\mathrm{out}}$ be the exact rows authorized by the effective coadd support policy. The MAP-owned family `SCI-MAP:uniform_observation_coadd_coefficient@1` assigns dimensionless $u_{op} = 1$ to every admitted row. It is equal-observation weighting, not precision. The coadd is:

$$\mathcal{S}^c_{\mathrm{out}} = \{p : \mathrm{authorize}_{\Pi^c_{\mathrm{eff}}}(p) = 1\}, \tag{30}$$

$$u_{op} = 1 \quad \text{for each admitted observation row,} \tag{31}$$

$$N^c_p = \sum_{o\in\mathcal{O}_p} u_{op}\widehat{m}_{op}, \qquad Q^c_p = \sum_{o\in\mathcal{O}_p} u_{op}, \tag{32}$$

$$\widehat{s}_p = N^c_p/Q^c_p, \qquad p \in \mathcal{S}^c_{\mathrm{out}}, \tag{33}$$

$$J^c_{\mathrm{out}}\boldsymbol{v}^c = (v^c_p)_{p\in\mathcal{S}^c_{\mathrm{out}}}, \tag{34}$$

$$\widehat{\boldsymbol{m}}^c_{\mathrm{out}} = (\widehat{s}_p)_{p\in\mathcal{S}^c_{\mathrm{out}}} = B_{\mathrm{out}}\widehat{\boldsymbol{m}}_{\mathrm{obs}}, \tag{35}$$

$$\widehat{k}^c_p = \frac{\sum_{o\in\mathcal{O}_p} u_{op}\widehat{k}_{op}}{Q^c_p}, \qquad p \in \mathcal{S}^c_{\mathrm{out}}. \tag{36}$$

$$C_{c,\mathrm{out}} = B_{\mathrm{out}}C_{\mathrm{obs}}B^{\mathsf{T}}_{\mathrm{out}} \quad \text{only when every required block is available,} \qquad R_{c,\mathrm{out}} = B_{\mathrm{out}}T_{\mathrm{obs}}. \tag{37}$$

Here $\widehat{\boldsymbol{m}}_{\mathrm{obs}}$ is the stacked signal on exact observation-row identities; B_{out} is the centered-integer equal- observation averaging operator; T_{obs} stacks exact compatible fixed-state observation response objects; and C_{obs} is the block covariance with every available within- and cross-observation block explicitly located. Equation (37) is a complete numerical covariance claim only when every block required by the exact covariance-qualified role is available. Otherwise the state is explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable. Unknown blocks are not zero or independence. The coadd operator B_{out} has rows only in $\mathcal{S}^c_{\mathrm{out}}$. Coadd accumulators or dense storage slots outside that set do not become a zero-valued scientific coadd. On its exact domain the coadd is a convex weighted mean. One admitted observation reproduces itself; two admitted observations produce their equal-observation mean. Its normalization is not precision. With cross-observation covariance, the exact conditional covariance retains those blocks on the same selected rows. A correlated-GLS benchmark may require negative combination coefficients, placing it outside this ordinary positive-coefficient contract.

The exact base-coadd role makes response advisory and unavailable-compatible; the response-bearing role requires compatible source domain, response basis, class, units, normalization, parent, and rows for every member. If any member response is unavailable or incompatible, the base role may retain the unchanged numerical signal membership but must report coadd response unavailable; it forms no hidden response subset. The covariance-qualified role similarly requires all named blocks. Retained exposure and admitted observation count follow the same atomic bundle admission, while exposure retains its unique-original own-coordinate placement. A rejected observation changes no coadd state anywhere, even if a subset of its planes would otherwise appear compatible.

7 Identity, WCS, units, and lifecycle

A map cannot be reconstructed from its collection position. Paired PTC x/r components remain upstream detector coordinates and are carried only when a MAP-facing fact requires their identity; MAP does not recast r as a response. The minimum identity includes the nonpolarimetric total-intensity-equivalent product role; external observation or coadd ID; array/group and occurrence domains; estimator, projection, PTC and coadd coefficient families; response/covariance states; signal unit and exact nominal-beam parent; complete AST frame/WCS; exposure convention; MAP profiles; exact PTC/AST generations and product identity. A STOKES field is only a locator unless separate quantity authority exists. The policy must be scientifically identifiable, durable, and comparable, but this contract prescribes no JSON, hash, or sidecar serialization. Array ID, name, index, network, map group, map index, detector occurrence, and container position remain distinct.

Science maps bind the complete frame identity supplied by `SCI-AST:rtc_output_grid_coordinates@1`, including the actual reference system, equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued axes. MAP never equates J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name/default. A Pointing/OOF ordinary-map use, if the owner registers it, uses a declared AltAz tangent plane in arcseconds; its mode-specific meaning is outside SCI-MAP. FITS/WCS coordinates are one-based and in-memory row/column indices are zero-based.

Full-precision typed or sidecar WCS is the lossless admission and provenance authority. A physical FITS serialization may differ by at most 0.1 arcsec in maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. This is a representation tolerance, not authority for a different grid.

Requested state records the immutable accepted intent. Effective state records the executable supported plan. Observation-resolved state replaces all observation-specific arrays, mappings, geometry, coefficients, and provenance. Realized state records what actually ran, all accumulators and distinct facts, products, offsets, representation states, cardinality, and failures. State flows one way; execution does not rewrite the request or inherit a prior observation's resolution.

8 Failure, downstream boundary, and immutable parentage

Failure semantics protect the scientific bundle from partial truth. An invalid candidate can be excluded at contribution scope; an unsupported pixel is MAP-local invalid; an incompatible observation is rejected atomically; an unrepresentable finite aggregate, coordinate/index conversion, missing required companion, or failed required publication propagates at its declared product scope before live state mutation. A completion marker cannot override a missing required output.

No downstream package receives a bare signal plane as a complete map. The boundary includes the estimator and normalization identity; response state; conditional covariance/formal-weight state and assumptions; distinct map facts; additive-bias and null-mode state; units; WCS; grouping; lifecycle; base/unfiltered MAP parentage; dependency versions; and failures. NOI may propagate a fixed operator but owns realization construction and empirical calibration. FLT owns filtered support, response, covariance, and validity. SRC, MODE, and BEAM own inference and fit uncertainty. FRUIT owns recurrence and iteration state.

Every later analysis is responsible for deciding which facts its own claim requires. An unavailable response prevents the original MAP bundle from making an unsupported response-dependent claim, but it does not prohibit analysis or invalidate other map claims. A significance analysis cannot relabel merely formal standardization; an absolute-level claim cannot ignore an unobservable offset. This contract defines no closed-world registry of all future uses. A downstream product may have local validity or eligibility, but it cannot rewrite MAP-local base-product validity or sever the immutable base/unfiltered MAP-parent identity.

When an admitted noise realization is described as using the same operator, membership, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and coadd admission are fixed or identically prescribed. Recomputing them from randomized data creates a different estimator, regardless of matching array shapes or filenames.

9 Uncertainty, validity, validation, and claim layers

The contract deliberately separates four reasoning layers:

1. **Uncertainty layer:** conditional covariance and response are propagated through the fixed operators; omitted state uncertainty remains explicit.
2. **Validity layer:** contribution eligibility, numerical support, science-policy support, and MAP-local base-product validity determine the exact rows on which the estimator is scientifically available; unsupported rows are absent, not zero-filled observations.
3. **Validation layer:** preregistered analytic fixtures, matrix comparisons, injections, covariance controls, representation round trips, and observational references test distinct claims.
4. **Readiness layer:** implementation evidence, operational reliability, data completeness, and release authority are assessed elsewhere.

Claim layer	Status here	Evidence needed before an achieved claim
Algebraic scientific contract	Proposed author draft	Scientific-owner review and contract freeze; analytic predictions are specified, not executed.
Engineering conformance	Not assessed	Requirement-by-requirement inspection and tests against one exact candidate identity.
Representation/response fidelity	Not assessed	WCS round trips and delta, template, extended, variable-coverage, and edge injections with preregistered bounds.
Observational performance	Not assessed	Independent astronomical amplitude, shape, position, covariance, and repeatability references where available.
Production readiness	Not assessed	Complete required artifacts, failure/log audit, operational gates, owner release decision, and no disqualifying gaps.

Floating reductions require a preregistered bound derived from operation count and conditioning. For a sum of n terms, a reference policy may use $\gamma_n = n\epsilon/(1-n\epsilon)$ and a fixed multiple such as the admitted core's $8\gamma_n \sum_i |a_i|$, frozen before observing results. Integer counts, identities, masks, support, offsets, and product cardinality remain exact. Tolerances are not relaxed after failure.

The contract's 25 predictions are listed formally in Appendix D. No validation run, implementation inspection, observational reduction, or production decision was performed during this scientific authorship stage.

10 Owner decisions, external hard gates, and route disposition

Eight MAP-local decisions remain open: (1) the physical rationale for the adopted support thresholds; (2) future threshold change authority and evidence; (3) the minimum response information that the original MAP product must provide for each response-dependent claim it makes; (4) the minimum covariance representation that must persist rather than remain lineage-resolvable; (5) whether physical observation-map publication is an abstract obligation beyond the effective v0.1 required-product rule; (6) whether Pointing/OOF ordinary arithmetic is a registered reuse of this operator; and (7) the admissible numeric domain, boundary-case dispositions, and failure behavior for the dimensionless $c =$ `coverage_cut`; and (8) canonical-grid preparation and future reprojection/mosaicking ownership. OD-008 remains a stable resolved identity: only `SCI-MAP:one_hot_containing_pixel@1` is authorized, with lower-inclusive/upper-exclusive cells, outer-upper-boundary loss, no wrap or clamp, and fractional projection deferred. SCI-MAP-CI-001 is also resolved.

Route level	Contract status	Exact remaining gate
Conditional estimator authority	Defined	Positive-coefficient observation and equal-observation centered-integer coadd equations are conditional scientific authority.
Source-closed numerical observation map	Unavailable	Exact PTC MAP-facing coefficient family and exact owner-admitted numerical `coverage_cut` state/value remain open; every required boundary/profile/source and realized same-n parent must also match.
Response-bearing science map	Conditional or unavailable	Requires an exact sufficient fixed-state, PTC full-procedure, or PTC+MAP re-resolved family for the named claim; whole-chain RTC-to-CAL-to-PTC-to-MAP response is unavailable; missing membership never creates a hidden subset.
Uncertainty-qualified map	Conditional or unavailable	Requires the exact covariance representation and every term/block needed by the named claim; unknown is not zero.
Compatible-grid coadd	Defined conditionally	Requires source-closed admitted observation bundles and `observation_coadd_admission@1`; response/covariance qualification remains role-specific.
Future reprojection/mosaic	Not authorized	OD-009 and a separate scientific contract remain required.

External hard gate	Contract disposition	Route effect
PTC MAP-facing coefficient	Open in PTC-OD-010	Blocks every ordinary numerical observation map.
Admitted numerical `coverage_cut` state	Open in MAP-OD-007	Blocks support-row construction and every numerical map.
`SCI-PTC_TO_SCI-MAP v0.1/r0.1`	Source-closed in this packet	Required exact digest/version; a mismatch blocks the route.
MAP upstream-admission profile	Exact manifest-bound Registry record	A missing or incompatible `SCI-MAP:map_upstream_admission@2` evaluation blocks candidate admission.
Exact AST same-n coordinate parent	Contract role closed; realization input-dependent	Missing/mismatched parent excludes the occurrence.

Original-footprint exposure lineage and AST ALIGN-grid parent	Exact boundary closed; realization input-dependent	Missing stable-original, target-WCS, validity, or boundary facts make required exposure unavailable and block a product that designates it required.
Response/covariance state	Claim-dependent	Does not by itself invalidate the base signal; blocks only a role whose exact policy requires stronger information.

The contract is coherent at this conditional boundary. It does not authorize an ordinary numerical realization while any hard structural gate remains open, and it establishes no implementation conformity, response fidelity, validation, observational performance, freeze, readiness, or production authorization.

A Owner-decision register

This compact register is generated from `SCIENTIFIC_OWNER_DECISION_LEDGER.md`. The mechanical contract checker requires exact decision IDs, open statuses, decision prompts, and conservative interim dispositions to match before either PDF is accepted. Rendering this register does not answer or close an item.

Decision/status	Exact decision requested	Conservative draft behavior pending decision
`SCI-MAP-OD-001` **OPEN**	What physical rationale is authoritative for the adopted v0.1 normalization-support and science-policy-support threshold rule?	State the formula exactly, label it adopted policy, and make no physical optimality, completeness, noise, or response claim from it.
`SCI-MAP-OD-002` **OPEN**	Who may change the support-threshold rule after v0.1, and what evidence is required for a successor?	Freeze the v0.1 formula and population identity; treat any altered multiplier, quantile/index, population, or predicate as a contract change.
`SCI-MAP-OD-003` **OPEN**	What response information must the original v0.1 MAP product provide for each response-dependent claim it makes, and which original product claims remain unavailable when response information is incomplete or unavailable?	Report the actual response state, meaning, domain, and limits; fabricate neither zero nor identity response. Make no unsupported response-dependent claim. Permit later response or corrected-map products only with new versioned identity bound to the original MAP product and processing identity.
`SCI-MAP-OD-004` **OPEN**	What minimum covariance representation must persist in a base/unfiltered MAP bundle, and what may remain resolvable through lineage?	Report what uncertainty/covariance information exists, with meaning, domain, assumptions, limitations, omissions, and lineage. Do not promote normalization, hits, exposure, or diagnostic weight. Permit later estimates as new versioned products bound to the original.
`SCI-MAP-OD-005` **OPEN**	Beyond the adopted v0.1 effective required-output rule, is physical observation-map publication an abstract contract obligation whenever coaddition is requested?	Preserve the v0.1 rule that every product designated required by the effective plan, including required observation products during coaddition, must publish successfully. Do not generalize this into a future universal storage mandate.
`SCI-MAP-OD-006` **OPEN**	Is Pointing/OOF ordinary-map arithmetic a registered use of the shared SCI-MAP v0.1 operator, with mode-specific interpretation remaining outside this package?	State only that reuse is conditional on a fully conforming input bundle; grant no mode-specific fitting, astrometric, response, or production authority.
`SCI-MAP-OD-007` **OPEN**	What numerical domain is authoritative for the dimensionless scalar `c=coverage_cut`, including the required disposition of negative, zero, non-finite, and greater-than-one values, and what failure behavior applies when a numerical state is not authorized?	Assert no general numerical domain. Require the exact numerical value to be explicitly admitted by the owner-authorized effective support policy; otherwise fail closed before constructing support-authorized output rows or mutating a required product.
`SCI-MAP-OD-008` **RESOLVED**	Which sample-to-pixel projection classes are authorized for ordinary SCI-MAP v0.1, what normalization applies to MAP-owned `G_pi`, and how is boundary loss represented?	Authorize only `SCI-MAP:one_hot_containing_pixel@1` with lower-inclusive, upper-exclusive cells; `G_pi=1` only for the unique finite in-grid containing pixel and zero elsewhere; outer-upper-boundary and out-of-grid loss; no wrap or clamp; fractional projection deferred.
`SCI-MAP-OD-009`		

Decision/status	Exact decision requested	Conservative draft behavior pending decision
OPEN	Is upstream crop or pad to a canonical common grid an authorized preparation for v0.1 observation coaddition, which package owns it, and which future package owns reprojection or mosaicking beyond centered-integer placement?	Reject every odd-difference or otherwise incompatible observation bundle. SCI-MAP performs no crop, pad, fractional shift, reprojection, interpolation, or mosaic; no other package is named as authorized until the owner decides.

B Requirement and crosswalk summary

The detailed 52-row requirement crosswalk and 25-row prediction crosswalk remain in `CROSSWALK.md`. This compact index makes the rationale's traceability visible without duplicating the normative clause text or the detailed row table. Appendix C renders every canonical requirement, and Appendix D renders every prediction, directly from the shared authority.

Authority range	Scientific surface	Principal rationale / open gate
REQ-001	Version and revision identity	Cover; owner freeze
REQ-002–010	Capability, input, identity, projection, coefficients, lifecycle, and admission	Sections 1, 7, 8; upstream producer facts
REQ-011–018	Ordinary accumulators, exact support-row publication, response, and absolute mode	Sections 1–3; OD-003
REQ-019–024	Row-selected covariance, formal weight, nuisance state, and statistical labels	Sections 2, 4; OD-004
REQ-025–035	Separate incidence/contribution/exposure/support facts, threshold policy, exact rows, and MAP-local validity	Section 5; OD-001, OD-002, OD-007
REQ-036–042	Complete bundles, atomic centered-integer coadd, row-selected coadd response/covariance, and GLS boundary	Sections 6, 8; OD-003, OD-004, OD-007
REQ-043–048	Product identity, WCS, provenance, failure, and publication	Sections 7–8; OD-005, OD-006
REQ-049–052	Immutable base MAP parent, consumer boundary, fixed realization operator, and claim separation	Sections 8–9; OD-003, OD-004, OD-006, OD-007
PRED-001–025	Analytic consequences and conformance/representation challenges	Sections 3, 9; Appendix D; not executed
OD-001–009	Stable scientific-owner decisions	Section 10; Appendix A; eight OPEN, OD-008 RESOLVED
SCI-MAP-CI-001	Resolved dimensionless classification and bounded amendment	Section 5; equations, REQ-031/032, and PRED-012 amended in place

C Canonical requirements

This appendix is rendered directly from the shared canonical LaTeX authority. Requirement identifiers are stable within SCI-MAP v0.1. Editing prose in this document does not silently change them.

SCI-MAP-REQ-001 – Versioned authority. The scientific contract version shall be v0.1. The rendered document revision shall be recorded separately as r0.7.1; changing layout or exposition alone shall not be represented as a new scientific contract version.

SCI-MAP-REQ-002 – Capability and realized PTC input. The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated-x product $z_i = Z_i^{\mathrm{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent `mJy/beam` convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

SCI-MAP-REQ-003 – Exact occurrence identity. Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n, exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

SCI-MAP-REQ-004 – Producer facts and MAP admission. SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/typed availability, and coefficient/QC proposition without reinterpretation. The PTC coefficient/QC proposition passes only when the coefficient is requested, applicable, eligible, and realized under the exact PTC-owned coefficient/QC profile for the declared MAP-facing family. It does not grant MAP use. SCI-MAP shall separately apply the source-bound occurrence profile `SCI-MAP:map_upstream_admission@2`; the manifest-bound SCI-VAL Registry revision binds and VAL Core evaluates that MAP-authored policy but does not author it. The immutable r0.5 profile `SCI-MAP:map_upstream_admission@1` is historical, is not a compatibility alias, and shall not replace `SCI-MAP:map_upstream_admission@2`. Every fact, decision, cause, scope, and immutable reason remains separately typed.

SCI-MAP-REQ-005 – Coordinate and projection authority. MAP shall consume `SCI-AST:rtc_output_grid_coordinates@1` for the same stable RTC sample n and complete parent chain as z_i; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through `SCI-MAP:one_hot_containing_pixel@1`: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

SCI-MAP-REQ-006 – Ordinary coefficient status. The ordinary route requires one exact PTC-owner-selected MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052–055 and `SCI-PTC_TO_SCI-MAP v0.1/r0.1`. No MAP-facing coefficient family is selected by frozen SCI-PTC v0.1/r0.5 or bound in the SCI-MAP r0.7.1 source packet. MAP does not select this PTC-owned family. Ordinary numerical MAP realization is therefore unavailable until that owner decision is versioned and bound. Availability establishes family and generation identity, index/broadcast relation, transformed-product compatibility, readable payload presence, typed availability, and cause; it does not require finiteness. MAP alone classifies an authorized payload as finite positive, exact zero, negative finite, non-finite, or unrepresentable. MAP shall infer neither unity, loading, sensitivity, scatter, nor precision; cross-generation use requires explicit compatibility.

SCI-MAP-REQ-007 – Exposure lineage and accounting. ALIGN-owned e_{acq} and valid-original e_{vo}, in seconds, shall remain immutable on stable original occurrences through RTC/CAL/PTC. The products are normatively `upstream_eligible_original_footprint_exposure` and `retained_original_footprint_exposure`. The shorter prose labels "upstream-eligible exposure" and "retained exposure" are explicit aliases for those products and no others. MAP shall deduplicate by exact observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation; place each positive-e_{vo} original exactly once through `SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1`, its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, and `SCI-MAP:one_hot_containing_pixel@1`; and never use a descendant RTC-output coordinate. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are required. Upstream-eligible and retained exposure differ only by ancestorpopulation restriction. Causal influence, representative RTC occurrence, and original-footprint exposure remain separate. These exposure products are not complete temporal support, effective map integration time, precision, or normalized-map influence. Missing original-coordinate authority makes the affected exposure role unavailable. Invalid-original has $e_{\mathrm{vo}} = 0$; synthesis, replacement, donor reuse, overlapping filters, and decimation create no exposure. Boundary loss contributes nowhere. Coadd exposure unions atomically admitted observation-scoped original identities.

SCI-MAP-REQ-008 – Response information state. MAP shall keep fixed-state linear response, PTC full-procedure finite differences, and PTC+MAP re-resolved procedure response as distinct typed families. An available fixed-state object shall bind source domain, PTC-sample codomain, units, parents, membership, support, coefficient state, response basis, and limitations; a full-procedure object shall additionally bind perturbation family α and a separately typed $\Delta\mathcal{S}_{\text{PTC-FP}}$ state-change record. The PTC+MAP family reruns PTC and resolves Π_0 and Π_+. A whole-chain RTC-to-CAL-to-PTC-to-MAP response is reserved for a separately authorized complete rerun and is unavailable and not claimed here. MAP shall otherwise report an explicit limited or unavailable state. Missing response shall not invalidate an otherwise valid base map, fabricate an identity, or prohibit later analysis. A later simulated response or corrected map is a new versioned product.

SCI-MAP-REQ-009 – Immutable plan and lifecycle. The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

SCI-MAP-REQ-010 – Factored contribution predicate. For each pixel, membership in $\mathcal{C}_p$ shall be the conjunction of ten exact Boolean pass predicates derived from separately retained typed records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition $\text{pass}_{\gamma,\text{QC}}$; MAP upstream admission; AST coordinate validity; finite z_i; finite strictly positive γ_i; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Every VAL-evaluated gate retains named request, applicability, eligibility, and realization fields; only request=`requested`, applicability=`applicable`, eligibility=`eligible`, and realization=`realized` passes. Every other field combination is nonpassing with its exact cause, failure scope, and immutable reason retained. A structural source- or profile-binding failure is recorded only when that structural condition actually occurs; an ordinary nonpassing field value is not thereby a structural failure. Evaluation follows the ordered DAG: A structural identity/parent, generation, source/profile, producer availability, and decision-realization checks; B retrieval only of structurally admitted payloads and MAP classification of z_i and γ_i; C AST validity, one-hot boundary/placement, and MAP-local required companions; D membership construction; E accumulation. All gates pass before payload arithmetic. Reading an authorized payload to classify it is not arithmetic, and excluded non-finite values never contaminate accumulation. T/F/U/C or four-field records shall never be conjoined directly; no numeric-zero mask represents exclusion.

SCI-MAP-REQ-011 – Ordinary accumulators. On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi}\gamma_i > 0$.

SCI-MAP-REQ-012 – Normalized publication. The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\widehat{m}_p = N_p/Q_p$, and an available response companion gives $\widehat{k}_p = K_p/Q_p$. No unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero.

SCI-MAP-REQ-013 – Accumulator interpretation. The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

SCI-MAP-REQ-014 – Constant preservation. With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

SCI-MAP-REQ-015 – Projection-specific estimand. The authorized base projection is exactly `SCI-MAP:one_hot_containing_pixel@1`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

SCI-MAP-REQ-016 – One exact MAP operator for linear companions. For one selected frozen plan Π, $A_{\mathrm{MAP},\Pi} \equiv A_{\mathrm{out}} \equiv D_{Q,\mathrm{out}}^{-1} J_{\mathrm{out}} G \Omega$ is the only MAP operator. Its identity binds the exact admitted PTC-sample input domain, support-authorized output-row domain, coefficient family/generation, one-hot projection plan, support policy, target WCS/grid, immutable parents, and lifecycle generation. Signal, fixed-state response, conditional covariance, an available realized PTC-grid response companion, fixed noise realizations, and a PTC full-procedure finite difference with MAP state fixed shall use that same operator, membership, placement, coefficient state, and output rows. A PTC-grid companion begins at MAP and receives the operator exactly once, never $H_{\mathrm{fixed},\Theta}$ again. Physical original-footprint exposure instead uses unique originals and each original's own AST ALIGN-grid coordinate; causal influence is not a linear exposure companion.

SCI-MAP-REQ-017 – Realized response claim. A stored kernel may be called a realized fixed-state response to template $\boldsymbol{\tau}$ only when its exact sample parent satisfies $\boldsymbol{k} = H_{\mathrm{fixed},\Theta}\boldsymbol{\tau}$ and the support-restricted map satisfies $\widehat{\boldsymbol{k}}_{\mathrm{out}} = A_{\mathrm{MAP},\Pi} H_{\mathrm{fixed},\Theta}\boldsymbol{\tau} = R_{\mathrm{fixed},\Theta}\boldsymbol{\tau}$ on exactly the signal rows. A PTC full-procedure finite difference $\Delta\boldsymbol{z}_{\text{PTC-FP}}$ plus its separate state-change record, and a PTC+MAP re-resolved difference shall retain their distinct family/domain identities and shall not be represented by that fixed matrix. Otherwise the response is a tracer, limited, or unavailable.

SCI-MAP-REQ-018 – Absolute-offset mode. SCI-MAP v0.1 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

SCI-MAP-REQ-019 – Conditional covariance. When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ, MAP shall propagate it as $C_{m,\mathrm{out}} = A_{\mathrm{MAP},\Pi} N_{\mathrm{PTC}} A_{\mathrm{MAP},\Pi}^{\mathsf{T}}$, retaining consequential off-diagonal terms on the exact operator domains. The equation is complete only on the declared available domain. Missing, limited, symbolic, summarized, lineage-resolvable, or unavailable terms remain typed and are never zero-filled or interpreted as independence; absence of a complete model does not invalidate the signal map.

SCI-MAP-REQ-020 – When normalization is precision. Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

SCI-MAP-REQ-021 – Formal diagonal weight. When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p, support-restricted precision $C_{m,\mathrm{out}}^{+}$, exposure, hits, and empirical statistical weight.

SCI-MAP-REQ-022 – Covariance and nuisance disclosure. The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

SCI-MAP-REQ-023 – Data-derived state. If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

SCI-MAP-REQ-024 – No significance promotion. Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. Empirical calibration remains NOI-owned.

SCI-MAP-REQ-025 – Distinct map facts. Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, `upstream_eligible_original_footprint_exposure`, `retained_original_footprint_exposure`, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

SCI-MAP-REQ-026 – Incidence layers. AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in $\mathcal{C}_p$. The one-hot touch convention counts the unique containing pixel only.

SCI-MAP-REQ-027 – Estimator contributions. Estimator-admitted contributions shall count exactly $i \in \mathcal{C}_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

SCI-MAP-REQ-028 – Admitted observation count. For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

SCI-MAP-REQ-029 – Upstream-eligible original-footprint exposure. MAP `upstream_eligible_original_footprint_exposure` shall union and deduplicate exact positive-e_{vo} original occurrences that are ancestors of route candidates admitted by `SCI-MAP:map_upstream_admission@2`. Each original is placed once by its own exact AST coordinate and the one-hot half-open rule; it is not placed at every descendant coordinate. The exposure population precedes coefficient positivity and MAP-local numerical rejection. Missing original-coordinate or lineage authority makes this exposure role unavailable.

SCI-MAP-REQ-030 – Retained original-footprint exposure. MAP `retained_original_footprint_exposure` shall use the same original-occurrence key and own-coordinate placement as REQ-029, restricted to originals that are ancestors of estimator-admitted members. Overlapping filters, decimation, replacement, synthesis, and multiple descendants do not duplicate seconds. Coadded exposure shall union observation-scoped original identities after atomic coadd admission; one original can occupy at most one pixel under the base one-hot family, or none after boundary loss. Neither exposure is precision.

SCI-MAP-REQ-031 – Adopted threshold selector. For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p; for count $N > 0$, select zero-based $k = \lfloor(\lfloor 0.75N\rfloor + N)/2\rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population identity and exact dimensionless support-policy scalar $c =$ `coverage_cut` shall be recorded. The effective support policy shall explicitly admit that exact numerical c state before support resolution; otherwise the required product shall fail closed.

SCI-MAP-REQ-032 – Separate support predicates. Only after the effective support policy explicitly admits the exact value of the dimensionless scalar c, numerical normalization support shall require an explicit finite positive $Q_p \geq Q_\star c/10$, and science-policy support shall independently require an explicit finite positive $Q_p \geq Q_\star c$. This contract asserts no general numeric domain for c. Every non-authorized numerical state fails closed before a support row set is constructed. Neither support state shall establish final validity or precision.

SCI-MAP-REQ-033 – MAP-local base-product validity. One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

SCI-MAP-REQ-034 – Classify before arithmetic. Only structurally admitted payloads shall be retrieved and safely classified for finiteness/sign before membership construction. All required gates shall pass before payload arithmetic. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

SCI-MAP-REQ-035 – Exceptional contribution states. Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

SCI-MAP-REQ-036 – Complete observation bundle. A base/unfiltered observation map shall be one immutable ordered bundle containing signal, numerator, normalization identity, response class/state, uncertainty/covariance state, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, lifecycle, exact PTC/AST parentage, and product identity. Limited or unavailable response/covariance states are complete disclosures, not automatic signal invalidity.

SCI-MAP-REQ-037 – Atomic coadd admission. A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

SCI-MAP-REQ-038 – Centered integer common grid. Observation and coadd shapes shall differ by nonnegative even row and column counts, with the corresponding integer offset mapping their reference pixels to the same world coordinate. Fractional shift, reprojection, interpolation, wrapping, implicit recentering, and a different sky grid are forbidden.

SCI-MAP-REQ-039 – Ordinary coadd estimator. The exact v0.1 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle eligible and realized under the VAL-governed aggregate profile `SCI-MAP:observation_coadd_admission@1`. It is an equal-observation coefficient, not precision or inverse variance. On exact centered-integer row identities, the coadd accumulates $N_p^c = \sum_o u_{op}\widehat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and publishes $\widehat{s}_p = N_p^c/Q_p^c$ only on $\mathcal{S}^c_{\text{out}}$.

SCI-MAP-REQ-040 – Coadd companions and unavailable response. The stacked signal vector $\widehat{\boldsymbol{m}}_{\text{obs}}$ shall carry explicit observation-row identities; B_{out} is the exact centered-integer equal-observation averaging operator; and T_{obs} is the stacked compatible fixed-state observation-response object on those same domains. Under the exact response-independent base-coadd role, response is advisory and unavailable-compatible: if any admitted member response is unavailable or basis-incompatible, the unchanged numerical signal membership may coadd only with coadd response typed unavailable, never a hidden subset or zero companion. A response-bearing coadd role instead requires every member response and exact compatible source-domain, basis, class, units, normalization, and row identity. Exposure and count follow the same atomic bundle admission, with exposure's unique-original placement.

SCI-MAP-REQ-041 – Coadd covariance and precision. C_{obs} shall be the block matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. Only when every block required by the exact covariance-qualified role is available may MAP publish the complete numerical equation $C_{c,\text{out}} = B_{\text{out}}C_{\text{obs}}B_{\text{out}}^{\mathsf{T}}$. Otherwise covariance shall be explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable; missing blocks are unknown, never zero or independence. The base signal coadd may remain valid under its named unavailable-compatible role. The uniform observation coefficient is not precision.

SCI-MAP-REQ-042 – Correlated-GLS boundary. A correlated minimum-variance solution using $K_p^{-1}\mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

SCI-MAP-REQ-043 – Minimum product identity. Every observation and coadd product shall identify its nonpolarimetric total-intensity-equivalent role; external observation/coadd, array/group, and occurrence domains; estimator and coefficient families; response/covariance states; signal unit and fixed-nominal-beam parent; complete AST frame/WCS; grid/projection/boundary; MAP profiles; exact PTC product/application and coefficient generations; exposure convention; lifecycle; parentage; causes; and product identity. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

SCI-MAP-REQ-044 – Frames and units. MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

SCI-MAP-REQ-045 – WCS and index persistence. Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

SCI-MAP-REQ-046 – Realized provenance. Realized state shall record requested, effective, observation-resolved, applied, and realized identities; exact PTC/AST/boundary/profile generations; every typed gate's named request, applicability, eligibility, and realization fields plus Boolean projection, immutable reason, cause, failure scope, and DAG stage; admitted contributions/bundles; the exact $A_{\text{MAP},\Pi}$ identity, accumulators, and row domains; unique-original exposure keys, stable ALIGN slots, layered AST ALIGN-grid coordinate parents, target WCS, boundary states, and separate influence edges; PTC λ, $\widehat{U}_{\text{corr}}$, $U_{\text{total},\Theta}$, removed-subspace and null-space identities; fixed-state, PTC full-procedure, and PTC+MAP re-resolved response family and state-change record; covariance state; offsets; required products; and failures.

SCI-MAP-REQ-047 – **Failure scope and atomicity.** An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

SCI-MAP-REQ-048 – **Required observation products.** A complete observation bundle remains required when coaddition is enabled, and every product designated required by the effective v0.1 plan shall be published successfully before completion is asserted. Whether future abstract contracts require physical observation-map publication independent of an effective product policy remains an owner decision.

SCI-MAP-REQ-049 – **Immutable base parent.** Downstream transformations shall preserve resolvable base/unfiltered MAP-bundle identity and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, or sever parentage. Later response or covariance estimates are new versioned products.

SCI-MAP-REQ-050 – **Complete consumer boundary.** No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include quantity and nominal-beam identity, estimator/normalization/coefficient/profile identity, honest response class/state and uncertainty/covariance disclosure, distinct map and exposure facts, PTC additive-reference/removed-component/null-space state, exact frame/WCS, shape/indexing, lifecycle, parentage, dependency versions, and failures. Later analyses choose their own claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility.

SCI-MAP-REQ-051 – **Same fixed noise operator.** An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

SCI-MAP-REQ-052 – **Method and claim separation.** This authority shall not be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

D Falsifiable predictions

These predictions are preregistered contract consequences or conformance challenges. Their presence is not validation evidence.

SCI-MAP-PRED-001 – Uniform constant. For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized MAP-valid output row; numerator and normalization scale together, and no unsupported scientific row is created. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-002 – Unequally weighted constant. For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-003 – One-pixel hand calculation. For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \widehat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-004 – One-hot boundary. A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-005 – Delta input. For a declared fixed state, a basis-pixel sky perturbation produces the corresponding support-row-restricted column of $R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta}$, including projection and boundary truncation; a PTC full-procedure or PTC+MAP re-resolved difference is separately typed and need not equal that column. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-006 – Point-source template. A declared point-source perturbation $\boldsymbol{\tau}_b$ produces $R_{\text{fixed},\Theta}\boldsymbol{\tau}_b$ only for the exact fixed response state and authorized rows; PTC full-procedure and PTC+MAP re-resolved families retain their perturbation/state parameters. The unavailable whole-chain RTC-to-CAL-to-PTC-to-MAP family is not inferred. Peak, integrated, and fitted amplitudes remain distinct functionals. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-007 – Resolved and Fourier inputs. At fixed state, resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{fixed},\Theta}\boldsymbol{\tau}$ on the authorized row set; a constant-input pass does not predict preservation of these modes, and procedure-response families are not inferred from the fixed matrix. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-008 – Variable coverage. Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-009 – Finite boundary. At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-010 – Coefficient states. Until an exact PTC-owner-selected MAP-facing coefficient family is bound, the ordinary numerical route is unavailable. Under a future exact family, structural availability is independent of numerical class; MAP alone classifies finite positive, exact zero, negative finite, non-finite, and unrepresentable. Zero is noncontributing and the other nonpositive/non-finite classes are invalid with exact cause; none is inferred as unity or precision. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states. A candidate whose typed gate record projects to Boolean nonmembership does not contaminate an accumulator. Structural exclusion prevents payload retrieval; structurally admitted payloads are read only for safe MAP numerical classification before membership and arithmetic. Its complete named-field or other typed record, immutable reason, and cause remain preserved; no universal producer-flag action is inferred. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-012 – Threshold selector. For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor(\lfloor 0.75N\rfloor + N)/2\rfloor$; empty input yields threshold zero, but no pixel gains support without its own finite positive coefficient. The dimensionless identity of c is fixed; any exact numerical c value not explicitly admitted by the owner-authorized effective policy fails closed before a support row set is constructed, and no numeric domain is inferred. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-013 – Missing response. A missing member response yields a typed unavailable response state on the complete signal membership, never a smaller hidden subset, zero, or identity. Fixed-state, PTC full-procedure, and PTC+MAP re-resolved response families remain distinct; whole-chain RTC-to-CAL-to-PTC-to-MAP response remains unavailable. The base MAP bundle remains usable for claims that do not require response; any later response or corrected map has a new versioned identity. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-014 – Atomic rejection. A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-015 – One-observation coadd. A coadd containing one admitted observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. It reproduces response only when the exact response companion is available; otherwise response remains typed unavailable. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-016 – Uniform observation coefficients. On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. The normalization is not precision. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-017 – Missing or invalid observation. An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-018 – Centered integer placement. Compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-019 – Cross-observation covariance. For two maps with variances σ_1^2, σ_2^2 and covariance c, a complete covariance-qualified uniform coadd includes c. If a required block is missing, covariance is partial, symbolic, summarized, lineage-resolvable, or unavailable rather than a complete numerical matrix; the missing block is never zero or independence, and equal-observation normalization is not precision. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-020 – Fixed noise operator. A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient $A_{\mathrm{MAP},\Pi} \equiv A_{\mathrm{out}}$ and B_{out} as the signal path; recomputed state exposes a named difference. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-021 – Sequential and parallel reduction. Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-022 – No false significance. A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-023 – No validity promotion. A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-024 – No ordinary predicate for JINC. A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.1. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-025 – Aggregate and index failure. An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion. *Primary claim layer: Engineering conformance.*

E Product vocabulary

Product	Unit	Contract meaning
Signal numerator	coefficient times `mJy/beam`	Pre-normalized accumulator; not a sky estimator.
Normalization	declared gridding unit	Positive denominator; precision only under Requirements 020–021.
Normalized signal	`mJy/beam`	Conditional ordinary map estimand whose vector contains exactly the effective-policy-authorized output rows; no unsupported zero-filled semantic extension.
Formal diagonal weight	inverse square signal unit	Conditional marginal inverse variance under named assumptions; not empirical significance.
Conditional covariance	square signal unit	Full fixed-state noise propagation on the exact scientific output-row domain, with omissions and representation status named.
Response/kernel	`mJy/beam` when unit-bearing	Realized template response only when the sample-parent identity is proven; otherwise tracer or unavailable.
Geometric/estimator hits	integer	Counts of declared populations, not coverage, precision, or validity.
`upstream_eligible_original_footprint_exposure`	named time/accounting unit	Unique-original acquisition-footprint accounting for route-candidate ancestors; explicit prose alias: upstream-eligible exposure; not precision.
`retained_original_footprint_exposure`	named time/accounting unit	Unique-original acquisition-footprint accounting for estimator-admitted ancestors; explicit prose alias: retained exposure; not precision.
Support facts	logical	Separate normalization and science-policy predicates; their effective conjunction defines output-row membership.
MAP-local base-product validity	logical	Authority for the base/unfiltered MAP bundle only; downstream uses own their eligibility.

F Admitted source register and independence statement

The scientific author used only the following admitted sources:

Source	Exact admitted identity
Scope Brief	`SCOPE_BRIEF.md` SHA-256 `e2a9eb51edb5956191813b4cdbd23866e875d52cdf89cd8b6c272988b4f26674`.
Supersession Cover	`AUTHOR_SUPERSESSION_COVER.md` SHA-256 `8ea283525f18199d9760c3f672d145d71f0db87b320ab2e88a5c6635ef3d4aa0`.
Frozen core	Exact Git object `c28f18ed089657dae278caba2d6d6d65c7ec72f4:doc/audits/packages/SCI-MAP-001_INDEPENDENT_CORE.tex` SHA-256 `13dd5922bd492e381afcc3b015284216dde1ccc2199ece3d070ee577c7324381`.
Conventions and ownership	`AUTHOR_CONVENTIONS_AND_OWNERSHIP.md` SHA-256 `2d478cb6c5e897308d19614b8b01663318744971850c67459f84c7ddcd57c5c9`.
Frozen adjacent authority	SCI-PTC v0.1/r0.5 digest `8357961a49272adc40e27a8aa9e760e0d01ff2419ae2c88a62c0f93c9f959e66`; SCI-CAL v0.1 science r0.5/engineering r0.4 digest `413426f49edf1249f751a05bb8c6e9fd907b11e8da0530fe2da39814885efb22`; SCI-AST v0.1/r0.3 manifest digest `b54b6013750540f28aad02339a60bf36078980dc53b132beab73069d66ef3601`; SCI-VAL Core v0.1/r0.3 digest `2fc3b3ad329fe3035d442b43d1e564a74fc86ab49f85f56e87322d8553fad9a6`.
PTC-to-MAP boundary	`SCI-PTC_TO_SCI-MAP v0.1/r0.1`; one byte-identical copy is installed in each PTC and MAP packet.
MAP r0.7.1 profiles	`SCI-MAP:map_upstream_admission@2` and `SCI-MAP:observation_coadd_admission@1`, registered in the exact manifest-bound SCI-VAL Registry revision.
Original-footprint coordinate	`SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1`, binding SCI-AST Equation `align-parent`, Equations `tangent-parent`–`pixel-parent`, and REQ-073/080/081.
Owner closure directive	SCI-MAP r0.7 directive, SHA-256 `f7747eea28710d524e12c818b872ac3fcc49f413271f83c0644ae129949a8c8c`; semantic changes in `CHANGE_LOG_R0.7.md`.

No implementation, test, validation, reduction, production-status record, MAP-002/MAP-003 evidence, or web source entered this authorship. The bounded handoff review supplied only the owner-authorized authority corrections recorded above. The frozen ordinary derivation was consolidated rather than rederived for novelty.